

Bachelor's Thesis

Voice-Cloning-Based Social Engineering

Voice-Cloning-basiertes Social Engineering

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Abstract

Modern voice-cloning systems can reproduce target-speaker characteristics from short reference recordings, weakening the evidential value of a familiar voice in telephone communication. Human-perception studies show that cloned speech can be accepted as belonging to the genuine speaker and can be difficult to distinguish from authentic recordings [1], [2]. At the same time, recent vishing research indicates that attack susceptibility is shaped more strongly by persuasive context and delivery than by perfect voice human-likeness [3]. The resulting security problem is therefore socio-technical rather than purely acoustic.

This thesis models voice-cloning-based call impersonation as an authorization-boundary threat. Its primary contribution is an analytical framework that integrates system decomposition, STRIDE, an attack tree, and a seven-stage interaction lifecycle. The model separates acoustic identity presentation from the protected decision that produces security impact. Two standardized scenarios operationalize different influence structures: hierarchical pressure in an executive-payment context and distress-oriented assistance in a retail-customer context. Truth-Default Theory and limited dual-process reasoning are used to explain how contextual consistency, suspicion triggers, time pressure, and prosocial expectations can shape verification behavior [4], [5].

A supplementary German-language technical validation tests only selected boundary conditions of the analytical model. XTTS-v2 produced recognizable neutral target-speaker characteristics under stable conditions, while end-to-end conversational latency and emotional prosody remained important constraints. RVC generated clear German speech but insufficient perceptual target identity; Seed-VC improved resemblance but exhibited content-integrity instability. These observations delimit the tested implementations rather than establish general performance claims.

The analysis concludes that synthetic speech is best treated as a paralinguistic identity signal embedded in a broader social-engineering process. Security impact arises when that signal survives contextual scrutiny and is permitted to influence a protected authorization decision. Consequently, resilient defense should reduce the decision value of subjective voice judgments and require independent, high-assurance authentication or approval for sensitive actions [6].

Contents

Abstract	I
Table of contents	II
1 Introduction	1
1.1 Context, Significance, and Contribution	1
1.2 Problem Statement	2
1.3 Research Questions	2
1.4 Scope and Exclusions	3
2 Background and Foundations	3
2.1 Social Engineering, Vishing, and Human Trust	3
2.2 Truth-Default, Heuristic Processing, and Suspicion	4
2.3 Voice Cloning and Speaker Identity	5
2.4 TTS-Assisted and Voice-Conversion Interaction	6
2.5 Conversational Timing and Paralinguistic Consistency	6
2.6 Security Interpretation	7
3 Related Work	7
3.1 Voice Cloning, Realism, and Trust	7
3.2 Familiarity, Prosody, and Speaker Identity	8
3.3 Vishing, Persuasion, and AI Automation	8
3.4 Detection and Authentication	9
3.5 Threat Modeling and Research Gap	10
4 Methodology	10
4.1 Research Design	10
4.2 Threat-Model Construction	10
4.3 Scenario-Analysis Method	11
4.4 Supplementary Technical Validation	11
4.5 Ethical and Evidential Boundaries	12
5 Threat Model and Attack-Stage Analysis	12
5.1 Model Scope, Actors, Assets, and Trust Boundaries	12

5.2	STRIDE Analysis	15
5.3	Attack Tree	16
5.4	Seven-Stage Lifecycle, Failure Conditions, and Controls	17
5.5	Model Synthesis	20
6	Scenario Analysis	21
6.1	Scenario Design and Literature Integration	21
6.2	Scenario 1: Executive-Identity Impersonation	22
6.2.1	Scenario setup	22
6.2.2	Influence structure and identity presentation	22
6.2.3	Authorization-boundary analysis	22
6.3	Scenario 2: Retail-Customer Impersonation	23
6.3.1	Scenario setup	23
6.3.2	Influence structure and identity presentation	24
6.3.3	Authorization-boundary analysis	24
6.4	Cross-Scenario Comparison	25
7	Supplementary Validation of Technical Boundary Conditions	26
7.1	Evaluation Design	26
7.2	XTTS-v2 Findings	26
7.3	Speech-to-Speech Findings	27
8	Discussion and Conclusion	27
8.1	Main Findings	27
8.2	Answers to the Research Questions	28
8.2.1	Main research question	28
8.2.2	Sub-question 1	28
8.2.3	Sub-question 2	29
8.2.4	Sub-question 3	29
8.2.5	Sub-question 4	29
8.3	Contribution and Defensive Implications	29
8.4	Limitations	30
8.5	Conclusion	31
	Bibliography	32
	Overview of AI technologies used	37
	List of Figures	38
	List of tables	39

1 Introduction

1.1 Context, Significance, and Contribution

Social engineering remains a persistent cybersecurity problem because organizational controls are implemented through people as well as software. An employee who is technically authorized to approve a payment, disclose information, or change an account state can become the point at which an otherwise protected process is altered. Voice communication is particularly relevant because identity, intent, urgency, and emotional state are assessed in real time and through multiple paralinguistic cues. Recent reviews characterize vishing as an interdisciplinary problem spanning deception, persuasion, speech generation, computational paralinguistics, and security intervention [7].

Voice cloning changes the evidential value of one of those cues: speaker identity. State-of-the-art systems can produce speech that listeners frequently accept as human or as belonging to the corresponding target identity [1], [2]. Controlled experiments further show that cloned voices can influence trust through perceived similarity and audio quality [8]. These findings do not imply that a cloned voice independently causes compliance. The most recent large-scale vishing experiment instead found caller persuasiveness to be a stronger predictor of self-reported compliance than voice human-likeness [3]. This distinction motivates the central premise of the thesis: the security significance of synthetic speech emerges from its interaction with context, social influence, and authorization procedure.

The primary contribution is therefore strictly analytical. The thesis develops a call-specific socio-technical threat model that separates identity presentation from authorization, makes the human decision point an explicit trust boundary, and combines an attack tree with a seven-stage lifecycle. The scenarios then compare two structurally different influence mechanisms: hierarchical pressure, which pushes a Decision-Maker toward immediate compliance, and distress-oriented influence, which draws the Decision-Maker toward assistance. The contribution is not the invention of another generic social-engineering lifecycle; existing research already models social engineering as a staged process [5]. The contribution is the integration of synthetic acoustic identity with protected organizational decisions in a voice-specific authorization model.

To keep the analytical model grounded in current operational reality, a supplementary qualitative technical validation tests a narrow set of boundary conditions: local synthesis latency, neutral target-speaker resemblance, emotional prosody, and speech-content stability. This validation is contextual support only. It does not constitute a benchmark study, a listener experiment, or an independent empirical contribution.

1.2 Problem Statement

The central security problem is not that synthetic audio exists, but that telephone communication often assigns more evidential weight to a familiar or authoritative voice than that signal can justify. A plausible caller ID, target-like voice, timely pretext, and knowledge of organizational context can jointly create an identity hypothesis even though none of those cues independently authenticates the caller. If the recipient then allows that informal judgment to influence a protected process, a paralinguistic signal has acquired authorization relevance.

Existing research is fragmented across several domains. Speech research measures naturalness, similarity, prosody, intelligibility, and latency; deepfake research evaluates human or automatic detection; social-engineering research examines persuasion and cognition; and security engineering models assets, trust boundaries, and attack paths. Systematic reviews note that social-engineering experiments often omit contextual and cognitive factors that determine whether laboratory findings transfer to real attacks [5]. The research problem is therefore to connect these domains without reducing the threat to a single acoustic metric.

1.3 Research Questions

The thesis is guided by the following main research question:

How does voice cloning alter perceived caller legitimacy in call-based social engineering, and under which technical, contextual, and human conditions can that identity signal influence a protected authorization decision?

The main question is addressed through four sub-questions:

1. What actors, assets, attacker goals, assumptions, and trust boundaries characterize voice-cloning-based telephone impersonation?
2. How can the threat be represented through logical prerequisites and distinct interaction stages without conflating identity presentation with security impact?

3. How do contextual alignment, authority, familiarity, distress, conversational timing, and suspicion triggers shape the two analyzed interaction patterns?
4. What technical and procedural limitations constrain the operational value of synthetic or converted speech in the modeled scenarios?

1.4 Scope and Exclusions

Scope is limited to human-to-human telephone impersonation using local TTS or voice conversion. Autonomous agents, other media, system compromise, voice-biometric attacks, and caller-ID implementation are excluded. Standardized personas and consented German test material are used; no real targets are contacted and no operational fraud guidance is provided.

2 Background and Foundations

2.1 Social Engineering, Vishing, and Human Trust

Social engineering is the intentional manipulation of people to obtain information, access, or actions that would not otherwise be granted. In cybersecurity, the mechanism is socio-technical: the target typically possesses legitimate authority, and the adversary seeks to influence how that authority is exercised. Vishing is the voice-channel form of this problem. Unlike asynchronous phishing, a telephone call creates immediate interaction, turn-taking obligations, and paralinguistic information that must be interpreted while the conversation continues [7].

Authority, distraction, social proof, similarity, and deception recur in empirical analyses of real-world vishing material. Jones et al. found authority, social proof, and distraction among the most widely used persuasion principles in a corpus of 86 vishing examples [9]. These mechanisms are not exploits in the software sense. They change the decision environment in which the recipient evaluates identity and request legitimacy. The same request can therefore carry different weight depending on who appears to make it, when it is made, and whether the surrounding context seems coherent.

Recent fraud intelligence supports this interpretation. The UK National Assessment Cen-

tre assesses generative AI as primarily enhancing existing fraud types rather than creating entirely new ones, including targeted payment-diversion fraud using deepfake audio and video [10]. Europol likewise treats synthetic-media capability as an enabler within broader serious and organized crime, while FinCEN has warned financial institutions about fraud schemes involving deepfake media [11], [12]. This thesis consequently treats synthetic voice as an enabling identity signal within an established social-engineering process rather than as a self-contained attack class.

2.2 Truth-Default, Heuristic Processing, and Suspicion

A useful psychological account of vishing is Truth-Default Theory. In a direct vishing experiment, Armstrong, Jones, and Siami Namin found that recipients can initially treat callers as honest and revise that judgment when sufficiently diagnostic triggers appear, such as highly sensitive requests or contextual inconsistencies [4]. This is important because it shifts the analytical question from “why would a rational employee believe a fake voice?” to “what maintains or disrupts the initial presumption of legitimacy before the protected decision is reached?”

Dual-process theories provide a secondary, more general explanation. Type 1 processing is relatively rapid and low-effort, whereas Type 2 processing is more deliberative and working-memory intensive [13]. Time pressure can increase reliance on framing and heuristic cues in some decision tasks [14], but it would be too strong to claim that urgency deterministically forces a recipient into a separate cognitive system. In this thesis, authority, urgency, and distress are therefore modeled as factors that may reduce deliberative opportunity or motivation, while Truth-Default Theory explains how suspicion is activated or suppressed during the interaction.

Recent cross-domain deception research provides additional support for treating reflective processing as a vulnerability modifier rather than a binary cognitive state. Sabo, Sarno, and Black examined deceptive emails, news headlines, and voicemails and found that deception-detection performance was related across modalities; cognitive reflection predicted detection across the tested domains [15]. The result does not establish a vishing-specific causal mechanism, but it strengthens a cautious interpretation: recipients differ in their tendency to scrutinize digitally mediated claims, and conditions that compress time or attention may matter because they reduce the opportunity to exercise that reflective capacity.

This distinction matters for security controls. The threat succeeds neither because the recipient is irrational nor because the synthetic voice is flawless. It succeeds when the in-

teraction remains sufficiently coherent that no strong suspicion trigger causes the recipient to invoke independent verification before a protected action is taken. The authorization boundary is therefore both organizational and cognitive: policy defines what verification is required, while the recipient decides whether to execute it.

2.3 Voice Cloning and Speaker Identity

Voice cloning refers to speech generation conditioned on the characteristics of a target speaker. Modern zero-shot TTS systems can generate target-like speech from short reference audio; XTTS, for example, was designed as a multilingual zero-shot TTS system trained across 16 languages [16]. Voice conversion takes a different route by transforming source speech toward a target timbre while attempting to preserve linguistic content and, often, source prosody.

Speaker identity is multi-dimensional. Timbre and speaker embeddings capture relatively stable characteristics, while rhythm, pitch movement, speech rate, hesitation, breathing, and emotional expression contribute dynamic information. Recent perception studies show that voice clones can be judged as similarly real to authentic voices [2], yet familiarity can make a clone easier to reject as the same person [17]. A 2026 study further found that mismatched prosody reduced cross-source same-speaker judgments compared with prosodically matched pairs [18]. The relevant security concept is therefore not “voice realism” as a single scalar property, but the consistency of identity cues with the listener’s expectations.

Automatic similarity measures capture only part of this construct. Carbonneau et al. show that widely used automatic speaker-verification embeddings emphasize relatively static characteristics such as timbre and pitch range while underrepresenting dynamic traits such as rhythm [19]. This matters when interpreting synthetic-speech evaluations: a high embedding similarity can coexist with perceptible differences in cadence, phrasing, or expressive behavior. Accordingly, this thesis distinguishes measurable speaker similarity from perceived identity consistency and avoids treating either as a direct proxy for attack success.

This also explains why familiarity is double-edged. Familiarity can strengthen the prior expectation that a known identity is calling, but it provides a richer internal reference against which atypical rhythm, wording, prosody, or emotional behavior can be noticed [17]. For an unfamiliar executive, broad timbral resemblance may be sufficient to support an identity hypothesis; for a close colleague, the same clone may fail because speaker-specific behavior is missing.

2.4 TTS-Assisted and Voice-Conversion Interaction

TTS-assisted interaction converts typed text into target-like speech. Its strength is controllable linguistic content and, in mature systems, relatively stable pronunciation and speaker characteristics. Its weakness in live calls is interaction latency: the operator must understand the incoming turn, decide what to say, formulate the text, synthesize the audio, and deliver it. A limited prepared-response buffer can cover predictable acknowledgements or repetition requests, but it cannot solve open-ended dialogue and was not validated in this study.

Speech-to-speech voice conversion potentially reduces text-entry overhead because the operator speaks directly. However, VC must balance several competing objectives: preserve linguistic content, remove source-speaker identity, introduce target identity, retain useful prosody, and maintain naturalness. Recent VC research explicitly identifies source-prosody leakage and difficulty reproducing expressive paralinguistic behavior as unresolved design challenges [20], [21]. These trade-offs are directly relevant to the supplementary RVC and Seed-VC observations in Chapter 7.

2.5 Conversational Timing and Paralinguistic Consistency

Natural conversation is temporally constrained. Turn-taking research emphasizes that speaker transitions are typically fast and sensitive to timing [22]. Experimental telecommunications work shows that added latency increases overlap and between-speaker silence even when participants do not explicitly recognize the delay [23]. Human-agent research similarly reports substantially longer response latencies for conversational agents than for human partners [24]. Consequently, synthesis speed alone is not equivalent to conversational responsiveness.

Environmental sound can influence how pauses and audio imperfections are interpreted, but its effect is conditional. Travel noise, a busy office, a child crying, or connection disturbances can make a claimed situation more coherent when they match the pretext; they can also become suspicion triggers when they are inconsistent, exaggerated, or repetitive. The thesis therefore treats environmental audio as contextual evidence rather than an independent persuasion mechanism.

2.6 Security Interpretation

The cybersecurity significance of voice cloning follows from the separation between identity evidence and authorization. A caller’s voice can make an identity claim more plausible, but a voice cannot transfer funds, disclose records, or modify an account. Security impact occurs only when a recipient or system authorizes a protected action. This is why the most important trust boundary in the thesis is the point at which informal identity evidence is either rejected, independently verified, or allowed to influence a protected organizational process.

This interpretation also determines the defensive direction. Human perception and deepfake detection can be useful layers, but they should not carry sole responsibility for high-impact authorization. NIST SP 800-63B-4 explicitly prohibits voice-based biometric comparison and requires phishing-resistant cryptographic authentication in higher-assurance contexts [6]. The stronger security objective is therefore to make subjective acoustic judgment non-decisive for sensitive actions.

3 Related Work

3.1 Voice Cloning, Realism, and Trust

Recent perceptual evidence establishes that modern voice cloning can produce convincing identity cues without implying universal deception. Barrington, Cooper, and Farid found that participants judged cloned and genuine voices as the same identity in approximately 80% of trials and correctly classified generated speech as AI only around 60% of the time [1]. Lavan et al. similarly found that voice clones could be perceived as comparably real to human recordings, although they did not observe a general hyperrealism effect [2]. These studies establish a weakened evidential value of acoustic familiarity, not a guaranteed attack-success rate.

Trust effects have also been measured directly. Schanke, Burtch, and Ray showed experimentally that voice cloning can increase behavioral trust, partly through perceived homophily, and that disclosure of AI use did not remove the effect [8]. Jaggy et al. independently found that self-similar voices can increase perceived trustworthiness and likability [25]. Together, these findings support treating target-like speech as a social

signal that can affect trust formation even when listeners are not making a formal authentication judgment.

3.2 Familiarity, Prosody, and Speaker Identity

A second research stream complicates the simple assumption that more familiarity always increases attack effectiveness. Rosi, Soopramanien, and McGettigan found that familiar listeners were more sensitive to differences between authentic and cloned voices, whereas unfamiliar listeners often rated clones similarly or more positively on social traits [17]. This suggests a dual effect: familiarity can strengthen an identity prior, but it also increases the number of speaker-specific cues available for anomaly detection.

Prosody appears particularly important. Bakkouche et al. identified speech rate, rhythm, local pitch control, intonation, and speaker-embedding similarity as correlates of perceived naturalness across ElevenLabs, StyleTTS-V2, and XTTS-V2 [26]. Related 2026 work showed that prosodic inconsistency reduces the probability that listeners classify AI and human speech as belonging to the same identity [18]. These findings reinforce a distinction that is central to this thesis and to the local validation: reproducing speaker identity is not equivalent to reproducing speaker state, style, or context-specific behavior.

Language should also be separated from model and reference effects. Choi, Schäfer, and Steinebach compared ten recent TTS and voice-conversion approaches across English and German and reported that TTS generally outperformed VC, while cross-language performance differences were comparatively modest and the chosen target reference materially affected quality [27]. This evidence is particularly relevant to the present German-only validation: it supports treating language as a scope limitation, but not as a sufficient explanation for the observed RVC or Seed-VC limitations.

3.3 Vishing, Persuasion, and AI Automation

Vishing research increasingly supports a socio-technical interpretation. Triantafyllopoulos et al. describe the field as a convergence of deception, persuasion, speech processing, generation, and detection, while emphasizing the lack of integrated research and data [7]. Burda, Allodi, and Zannone reach a related conclusion across social engineering more broadly: a systematic review of 169 studies and 735 hypotheses found that experimental research often underrepresents context, cognitive processes, pretext variation, and targetization [5].

The strongest recent evidence comes from Heiding et al., who evaluated six leading voice systems in a 4,100-participant U.S. experiment. Self-reported willingness or uncertainty

about compliance averaged 16.5% across five scam categories and reached 36.1% in a relative-in-distress scenario. More importantly, caller persuasiveness was the strongest predictor of compliance, exceeding voice human-likeness as an explanatory factor [3]. This result directly supports the thesis’s emphasis on trust structure and interaction design rather than acoustic perfection.

Evidence is also beginning to extend beyond English-language experiments. Albayrak evaluated Turkish synthetic vishing material with 394 participants and reported very high rates of fake content being perceived as real; official-institution impersonations were the most convincing category in that study [28]. The experimental design and generation pipeline differ substantially from the present thesis, so its percentages cannot be transferred to the scenarios below. Its relevance is narrower but important: language alone does not define the social-engineering risk, and institutional framing can materially shape perceived legitimacy in non-English settings.

Automation itself is no longer hypothetical. Figueiredo et al. demonstrated a fully AI-automated vishing architecture combining an LLM, speech recognition, and TTS in a controlled 240-participant experiment [29]. Heiding et al. further argue that the immediate risk of current AI vishing may lie in economics and scalability rather than “super-human” persuasion [3]. Because this thesis excludes autonomous agents, its contribution is deliberately narrower: it isolates how synthetic target-speaker identity interacts with a human authorization decision in a human-operated call.

3.4 Detection and Authentication

Audio-deepfake detection remains an active defensive field. ASVspoof 5 expanded prior challenge conditions to approximately 2,000 speakers and 32 attack algorithms, including contemporary TTS, VC, and adversarial attacks [30]. The 2026 challenge evaluation reports submissions from 53 teams and shows that performance degrades under adversarial manipulation and neural encoding or compression [31]. These results do not imply that deepfake detection is ineffective; they show that robustness depends on attack and channel conditions.

For high-impact processes, authentication architecture is therefore more reliable than acoustic judgment alone. NIST’s current guidance states that biometric comparison based on voice shall not be used and defines phishing resistance in terms of cryptographic protocols that do not rely on user vigilance [6]. This supports a central defensive implication of the threat model: an organization should remain secure even when the Decision-Maker incorrectly believes the caller’s voice is genuine.

3.5 Threat Modeling and Research Gap

STRIDE and attack trees connect security effects to logical prerequisites [32], [33]. Novelty is bounded: the 2026 Synthetic Trust Attack Model already frames generative social engineering at the victim’s decision layer [34], and staged frameworks predate this thesis. The contribution is a call-specific integration of public audio exposure, synthetic identity, contrasting influence structures, independent verification, and protected authorization.

4 Methodology

4.1 Research Design

The study uses a multi-method analytical design. The primary method is structured socio-technical threat analysis informed by a targeted synthesis of recent literature on vishing, voice cloning, human perception, conversational timing, and security controls. The threat model is then operationalized through two standardized scenarios. A bounded local technical validation is added only to test selected feasibility assumptions.

The literature synthesis is not presented as a formal systematic review. Sources were selected from peer-reviewed speech, HCI, psychology, and cybersecurity venues; primary technical papers for the evaluated models; and official fraud or authentication guidance. Recent preprints are used only where they materially affect novelty positioning or identify emerging research directions, and are clearly treated as less mature evidence.

4.2 Threat-Model Construction

The model was constructed in five steps. First, the protected processes and assets were defined. Second, actors and trust boundaries were identified. Third, STRIDE categories were adapted to the socio-technical context without treating human persuasion as a software privilege escalation. Fourth, an attack tree was used to represent logical prerequisites and alternatives. Fifth, a seven-stage lifecycle ordered those prerequisites over time and made failure points explicit.

STRIDE is used as a categorization aid rather than a complete human-factor theory. Spoofing captures false caller identity; tampering captures unauthorized state changes;

information disclosure captures release of protected information; and elevation of privilege is interpreted narrowly as an authorization-equivalent effect achieved by inducing an authorized employee to act for an illegitimate caller. Repudiation and denial of service are secondary to the scenarios

4.3 Scenario-Analysis Method

Two scenarios were selected to test whether the same technical identity signal behaves differently under contrasting trust structures. Each scenario is evaluated against the same eight analytical dimensions:

1. Protected process and decision authority
2. Claimed identity and trust relationship
3. Reference-audio assumptions
4. Identity-presentation conditions
5. Dominant influence mechanism
6. Truth-Default and suspicion triggers
7. Independent verification path
8. Success and failure conditions

The scenario analysis is descriptive and comparative. It does not estimate attack probability. Its purpose is to determine which conditions must align before a synthetic voice can become security-relevant and where organizational controls can break the chain.

4.4 Supplementary Technical Validation

The technical component is deliberately subordinate to the analytical model. XTTS-v2 was used for German-language TTS cloning, while RVC v2 and Seed-VC were explored as speech-to-speech alternatives. The validation assessed local generation time, neutral target-speaker resemblance, emotional/prosodic limitations, and content stability. It did not compare commercial systems, perform a formal Mean Opinion Score study, or test deception against human participants.

4.5 Ethical and Evidential Boundaries

All technical tests used private, consented German-language material. No scenario identity was cloned, no real employee or organization was contacted, and no attack was deployed. The thesis deliberately avoids operational fraud scripts and implementation guidance for caller-ID manipulation. Claims derived from the local validation are confined to the tested configurations and are not generalized to the underlying technologies as a whole.

5 Threat Model and Attack-Stage Analysis

This chapter develops the central analytical model of the thesis. Its purpose is to explain how synthetic target-speaker identity can become security-relevant within a call-based social-engineering interaction without treating voice similarity itself as equivalent to compromise. The analysis combines complementary perspectives rather than competing models: the system-context view identifies where trust changes; STRIDE classifies the security properties at risk; the attack tree represents the logical conditions required for compromise; the lifecycle orders those conditions temporally; and the control analysis identifies where the attack path can be interrupted.

5.1 Model Scope, Actors, Assets, and Trust Boundaries

The generic threat model distinguishes active actors, impersonated identities, and the organizational environment in which the protected decision occurs. **Threat Actor A (TA)** represents an external adversary attempting to influence a protected organizational decision through voice-based impersonation. The **Impersonated Identity (II)** represents the person whose identity is presented by TA and is instantiated in Chapter 6 as either the **Executive Identity (EI)** or the **Retail Customer Identity (RCI)**. The **Decision-Maker (DM)** is an organizational actor with legitimate authority to influence or execute a protected process, such as approving a payment, changing a transaction state, or re-

leasing protected information. **Organization X (OX)** represents the organizational and technical environment in which these processes are governed by authentication, approval, logging, and authorization mechanisms.

The principal protected assets are organizational funds and payment state, confidential organizational information, customer and account information, transaction and customer records, approval and authentication workflows, and audit information. Their relevant security properties are primarily confidentiality, integrity, authorization assurance, accountability, and non-repudiation. The corresponding adverse outcomes are unauthorized payment or transaction-state modification, unauthorized disclosure of protected information, or another protected process change performed without the authorization evidence normally required for that action.

Authority, urgency, familiarity, distress, contextual alignment, and emotional presentation are not assets or security outcomes. They are *influence mechanisms*: factors that may alter how the DM interprets identity, legitimacy, verification requirements, or the perceived consequences of delaying or refusing a request. The model likewise distinguishes attacker capabilities from prerequisites and existing controls, as summarized in Table 1.

Tab. 1 Core elements of the threat model.

Category	Definition
Protected assets	Organizational funds and payment state; confidential organizational information; customer and account information; transaction and customer records; approval and authentication workflows; audit and attribution information.
Security properties	Confidentiality, integrity, authorized modification, authorization assurance, accountability, and non-repudiation.
Adverse outcomes	Unauthorized payment or transaction-state modification; unauthorized information disclosure; or another protected process change performed without valid authorization evidence.
Attacker capabilities	Collection of public contextual information; access to target-speaker audio; access to synthetic-speech capability; ability to initiate telephone contact; and ability to present a plausible caller identity.
Attack prerequisites	A relevant DM can be reached; the impersonated identity is plausible in the interaction context; suitable reference material is available; and the requested action falls within or near the DM’s authority.
Existing controls	Independent authentication, authenticated approval channels, role separation, dual control, transaction review, or equivalent organizational verification.
Out of scope	Direct system compromise, autonomous LLM agents, video deepfakes, voice-biometric attacks, and implementation of caller-ID spoofing.

The model does not depend on a particular TTS or voice-conversion implementation. The local systems evaluated in Chapter 7 provide only supplementary evidence about selected

technical boundary conditions.

Three security-relevant boundaries structure the model. The **identity-presentation boundary (B1)** is crossed when attacker-controlled signals enter the recipient’s decision environment. These signals may include synthetic speech, caller-ID metadata, contextual claims, timing, and environmental audio. None independently authenticates the caller; together they contribute to an identity hypothesis.

The **human authorization boundary (B2)** is the central analytical boundary. At this point, the DM determines whether the caller’s apparent identity and request justify continuation, independent verification, refusal, or escalation. Synthetic speech becomes security-relevant here because a paralinguistic identity signal can affect how the legitimacy of the caller and request is interpreted.

The **protected-process boundary (B3)** separates the human decision from the organizational action that creates material impact. A payment, disclosure, or account-state modification should cross this boundary only after the required authorization evidence has been established. A supporting **information-provenance interface** additionally describes how legitimate public information and public recordings can be transformed into adversarial preparation material.

These distinctions establish a deliberate separation between *identity evidence* and *authorization evidence*. A target-like voice may contribute to the former, but it should not constitute the latter.

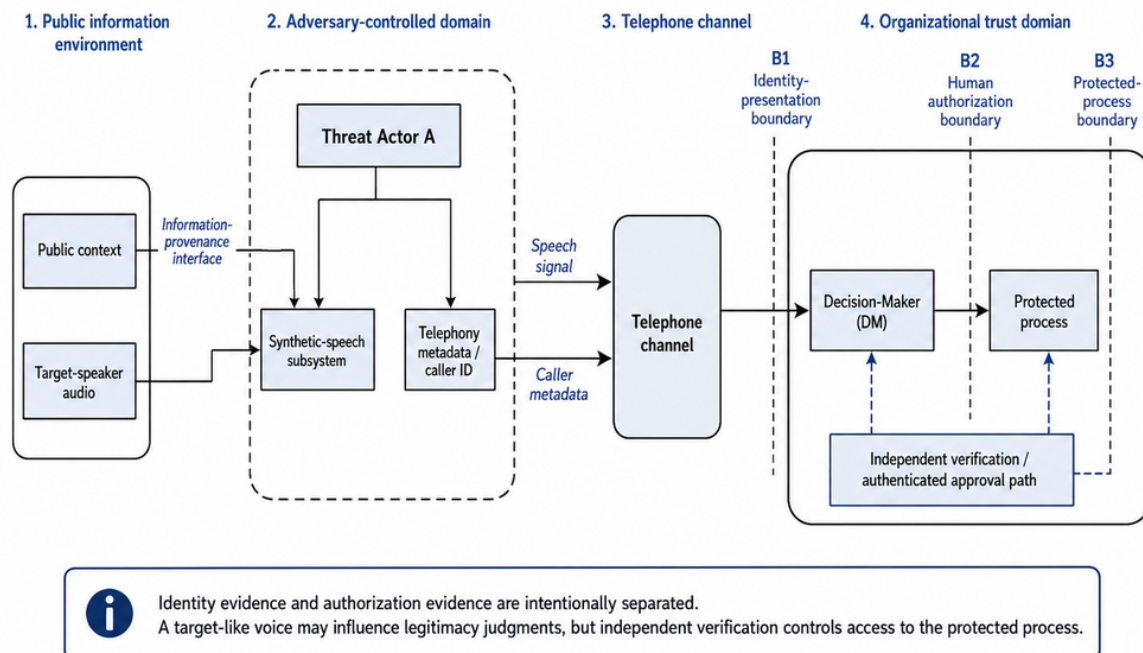


Fig. 1 System context and principal trust boundaries.

Figure 1 also distinguishes the incoming telephone interaction from an independent verification or authenticated approval path. Such evidence must originate from a channel whose trust does not depend on the incoming call, for example an independently initiated callback using trusted contact information, authenticated in-app confirmation, a second authorized approver, or another pre-established authentication mechanism. This allows the protected process to remain secure even when the DM initially accepts the caller’s voice as genuine.

5.2 STRIDE Analysis

STRIDE is used as a categorization aid for security-relevant components and data flows rather than as a complete theory of human manipulation. In its conventional form, STRIDE distinguishes Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege [32]. Because the present model combines technical and human elements, categories are applied only where their security meaning remains defensible.

Tab. 2 STRIDE categories and their application in the threat model.

Category	Property	Application
Spoofing	Authentication	TA falsely presents as the II through target-like speech and supporting identity cues. This is the primary STRIDE category at B1.
Tampering	Integrity	A successful interaction may result in unauthorized modification of payment, account, transaction, or workflow state.
Repudiation	Accountability	A secondary risk arises where audit evidence is insufficient to establish who initiated, requested, authenticated, or approved a protected action.
Information Disclosure	Confidentiality	Protected organizational, customer, or account information may be released to an illegitimate caller.
Denial of Service	Availability	DoS is not materially represented in the analyzed scenarios and therefore lies outside the primary threat path.
Elevation of Privilege	Authorization	No conventional technical privilege escalation occurs. The relevant analogue is induced authorization abuse, in which an already authorized DM exercises legitimate authority on behalf of an illegitimate caller.

The Elevation-of-Privilege category requires particular qualification. TA does not acquire the DM’s account, token, or technical privileges. Instead, the adversary attempts to cause an authorized principal to exercise existing authority for an unauthorized purpose.

The model therefore treats Spoofing, Tampering, and Information Disclosure as directly applicable; Repudiation as secondary; Denial of Service as outside the principal scenarios; and Elevation of Privilege only as an authorization-related analogue rather than a conventional privilege-escalation exploit.

5.3 Attack Tree

The attack tree represents logical dependence rather than temporal order. Following the attack-tree principle that lower-level conditions contribute to a defined attacker objective [33], the root node is formulated as an adverse security state:

Root goal: Cause an unauthorized protected action through voice-based impersonation.

Five high-level conditions must align for this goal to be reached and therefore have **AND semantics**: a DM with relevant authority must be reached; the claimed identity must become sufficiently plausible; the request, timing, and context must remain coherent; interaction credibility must be maintained until the authorization decision; and independent verification must not block the requested action. This formulation is deliberately stricter than defining attack success merely as “influence”, because influence without an unauthorized security outcome does not satisfy the root goal.

Within the identity branch, target-like synthetic speech is required by the scope of the model, while caller-ID plausibility, contextual knowledge, prior familiarity, and other identity cues can reinforce the claim without independently authenticating it. Within the interaction branch, TTS-assisted interaction and speech-to-speech voice conversion represent **OR alternatives**: either may provide the synthetic speech component, provided the selected method preserves sufficient timing, linguistic content, speaker characteristics, and delivery for the interaction to remain plausible.

The verification branch represents a defensive dependency. TA does not need to technically defeat every organizational control; the root goal may become reachable when independent verification is not triggered, is inadequately executed, or fails to prevent the protected action. Conversely, a mandatory independent approval mechanism can terminate the attack even where all preceding identity and contextual conditions are satisfied.

The tree is interpreted qualitatively rather than probabilistically. It establishes dependency and identifies possible failure points; it does not estimate the likelihood that any branch will succeed. Most importantly, it demonstrates formally that target-speaker similarity alone cannot satisfy the root condition.

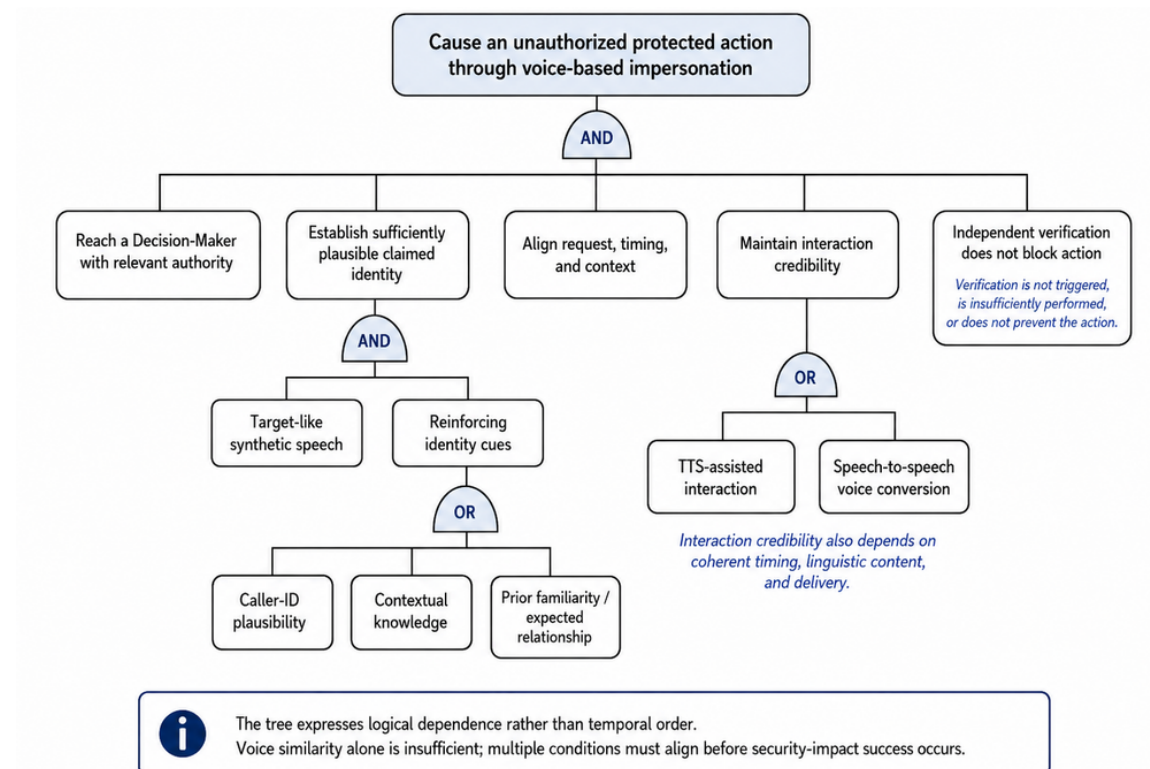


Fig. 2 Attack tree for unauthorized protected action through voice-based impersonation.

5.4 Seven-Stage Lifecycle, Failure Conditions, and Controls

Where the attack tree represents logical dependence, the lifecycle represents temporal development. The seven stages are not proposed as a universal social-engineering taxonomy; they provide a voice-specific ordering of the conditions identified above.

Stage 1 – Target and Decision-Point Selection. TA identifies a protected process and a DM whose role intersects with the intended security outcome. Reaching an employee without relevant authority does not produce direct security impact even if the impersonation itself is convincing.

Stage 2 – Context Acquisition. Information concerning organizational roles, relationships, timing, travel, transaction status, or other situational circumstances is collected to reduce inconsistency between the caller’s claim and the DM’s expectations. Context contributes plausibility but does not authenticate identity.

Stage 3 – Reference-Audio Acquisition and Assessment. Target-speaker material is obtained and assessed for quality, linguistic coverage, consistency, and suitability for synthesis or conversion. Public audio availability is therefore distinguished from operationally useful reference material.

Stage 4 – Interaction Preparation. The claimed identity, request, contextual explanation, delivery method, and expected verification points are aligned. The relevant analytical condition is coherence: identity, request, timing, and circumstances must not obviously contradict one another.

Stage 5 – Identity Presentation and Call Establishment. Attacker-controlled cues cross B1 and enter the DM’s decision environment. Target-like speech and supporting contextual or telephony cues create an initial identity hypothesis. Success at this stage is limited to *identity success*: the caller is considered sufficiently plausible for the interaction to continue. This does not constitute compromise.

Stage 6 – Credibility Maintenance, Influence, and Verification. The interaction progresses toward the protected decision. Truth-Default Theory is relevant because a provisionally accepted caller may continue to be treated as legitimate while the interaction remains coherent [4]. Sensitive requests, contradictions, unexpected behavior, speaker-specific inconsistencies, or policy requirements can instead act as suspicion triggers and initiate verification.

Influence mechanisms operate within this stage. Authority, urgency, familiarity, distress, or service expectations can alter the perceived cost of delay, refusal, or verification without deterministically causing compliance. Familiarity is particularly non-monotonic: it may strengthen the initial identity hypothesis while simultaneously providing additional speaker-specific expectations against which anomalous wording, rhythm, prosody, or emotional behavior can be detected [17]. Success at this stage is therefore *interaction success*: plausibility survives the decision-relevant exchange.

Stage 7 – Authorization and Security Outcome. The request is ultimately verified, rejected, escalated, or translated into a protected action. A third level, *security-impact success*, occurs only when a protected process is altered without the authorization evidence normally required for that action. Identity success, interaction success, and security-impact success are therefore analytically distinct: a highly convincing clone can satisfy the first two and still fail completely at the authorization boundary.

The lifecycle is deliberately failure-oriented. Contradictory context, unsuitable reference material, poor resemblance, implausible prosody, excessive delay, content instability, unexpected questions, or independently initiated verification can terminate the path before material impact. These failure conditions correspond to defensive controls at different layers, as summarized in Table 3.

The controls in Table 3 operate at different boundaries and should not be treated as interchangeable. Perceptual awareness and synthetic-speech detection primarily affect B1 by increasing suspicion. Organizational policy acts at B2 by determining when informal caller legitimacy must be replaced by independent verification. The strongest controls act

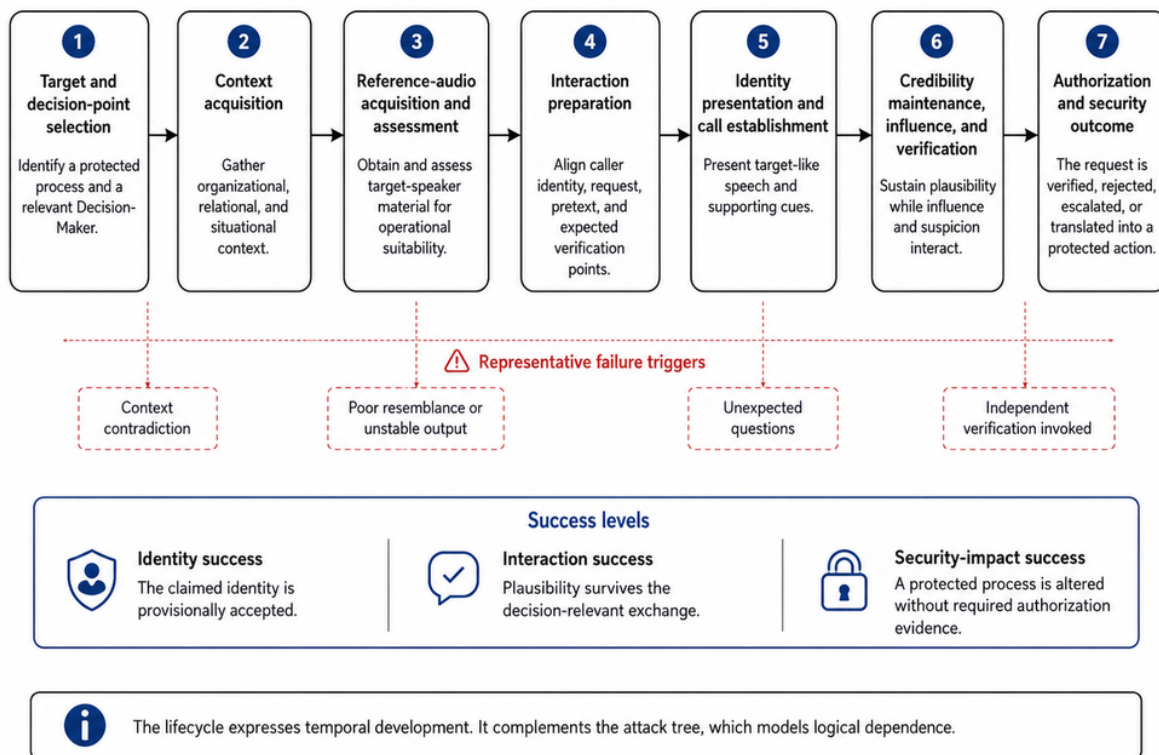


Fig. 3 Seven-stage lifecycle of voice-cloning-based telephone impersonation.

Tab. 3 Representative failure conditions and defensive controls across the lifecycle.

Location	Failure condition	Defensive control
Target selection	Wrong recipient or insufficient decision authority	Role separation; least privilege
Context acquisition	Claimed circumstances conflict with known information	Context verification; organizational awareness
Audio / preparation	Inadequate reference material or inconsistent identity characteristics	Upstream technical limitation
Identity presentation	Poor resemblance, implausible caller metadata, or inconsistent environmental cues	Human suspicion; optional anomaly or deepfake detection
Interaction	Excessive delay, content instability, unexpected questions, or implausible behavior	Challenge, escalation, or interruption of the interaction
Verification	Caller cannot satisfy independently initiated verification	Trusted callback; authenticated application; account-specific verification
Authorization	Required independent approval is absent	Dual control; step-up authentication; authenticated approval channel

at B3 because they make the protected action dependent on independent authorization evidence rather than on perceived caller authenticity. Independently initiated callbacks using trusted contact information, authenticated in-app confirmation, dual authorization, and phishing-resistant authentication are examples of such controls [6].

This distinction allows the organization to remain secure even when the DM incorrectly judges a synthetic voice to be genuine. The objective is therefore not to make employees perfect deepfake detectors, but to ensure that a false-positive identity judgment cannot independently produce material impact.

5.5 Model Synthesis

The combined model produces three principal analytical conclusions. First, **identity presentation and authorization are distinct security states**. Synthetic speech primarily affects the former: it may increase the plausibility of a claimed caller, but it cannot itself approve a payment, disclose information, or modify an account. Material impact requires a separate transition through the human authorization and protected-process boundaries.

Second, **security impact is conjunctive rather than purely acoustic**. A relevant DM must be reachable, identity and context must remain plausible, interaction credibility must survive the exchange, and independent verification must fail to block the requested action. High acoustic fidelity is therefore not sufficient for compromise and need not be the dominant determinant of the final outcome. The attack tree expresses this logical dependency, while the lifecycle shows how the corresponding conditions accumulate and may fail over time.

Third, **authorization-layer controls are more robust than perception-layer certainty**. Awareness, familiarity, and synthetic-speech detection can generate suspicion and reduce exposure, but they remain dependent on the ability to interpret the incoming interaction correctly. A stronger security design makes subjective caller legitimacy insufficient authorization evidence. When an independently authenticated approval path remains mandatory, false acceptance at the identity-presentation boundary does not automatically become security-impact success.

The complete model can therefore be summarized as:

$$\begin{aligned} & \textit{public context and target-speaker material} \rightarrow \textit{synthetic identity presentation} \\ & \rightarrow \textit{provisional legitimacy} \rightarrow \textit{influence and suspicion} \rightarrow \textit{independent} \\ & \textit{verification} \rightarrow \textit{protected authorization outcome} \end{aligned}$$

Voice cloning is positioned upstream of the protected decision. Its security relevance lies

in its ability to alter the identity evidence available to the DM; the eventual outcome is determined by the interaction between human judgment and organizational authorization controls. This separation provides the analytical basis for Chapter 6, where the same protected-decision architecture is examined under two different influence structures: hierarchical pressure in the executive scenario and distress-oriented assistance combined with limited prior familiarity in the retail-customer scenario.

6 Scenario Analysis

6.1 Scenario Design and Literature Integration

The scenarios operationalize the gap identified by recent vishing and social-engineering research: technical speech properties must be connected to psychological and organizational conditions [5], [7]. They also reflect the large-scale finding that persuasiveness and scam context can predict susceptibility more strongly than voice human-likeness [3]. The comparison therefore holds the protected-decision logic constant while varying the relationship through which influence is exerted.

Scenario 1 models pressure-oriented influence through hierarchy and urgency. Scenario 2 models distress-oriented influence through vulnerability and service expectations. Neither scenario assumes that these mechanisms deterministically override rational behavior. Instead, they are analyzed as conditions that may maintain the caller’s truth-default, reduce deliberative opportunity, or alter the perceived cost of refusing the request. The decisive variable remains whether independent verification is executed.

The scenario comparison is also informed by evidence that the social identity attached to a call matters. Authority is prominent in analyses of real vishing material [9], and recent Turkish user testing found official-institution impersonation particularly convincing within its own experimental setting [28]. These studies do not validate the specific scenarios used here; rather, they justify treating relationship structure and institutional role as first-order variables instead of decorative pretext details. The analysis therefore asks how the same acoustic identity technology interacts with two different expectations of the DM: obedience to senior authority versus assistance to a legitimate customer.

6.2 Scenario 1: Executive-Identity Impersonation

6.2.1 Scenario setup

Organization X employs a Decision-Maker in payment control. A transaction has been stopped or flagged because it requires additional authorization. Threat Actor A presents as the Executive Identity, a senior C-suite figure whose public profile provides a substantial voice footprint from social media, interviews, press conferences, public appearances, and news coverage. The executive is plausibly traveling or attending external meetings, creating a context in which a brief telephone request is not inherently anomalous.

The normal process requires written or secondary approval. The attack objective is not to defeat that control technically; it is to influence the DM into treating the caller's apparent identity and urgency as sufficient reason to make an exception.

6.2.2 Influence structure and identity presentation

The dominant trust relationship is hierarchical authority. The Executive Identity is presented as impatient, forceful, and unwilling to accept procedural delay. The cloned voice functions as a paralinguistic identity cue that makes the authority claim more plausible, while the pretext explains why the executive is calling directly and why time appears limited. Public formal recordings may provide substantial target-speaker material, but they may not reproduce the private, concise, or irritated speaking style expected in an internal call. This creates a natural gap between speaker identity and speaker state.

The social mechanism is pressure-oriented. Authority and time pressure can reduce the opportunity for reflective verification, while the caller's impatience increases the perceived professional cost of delay. This should not be described as a deterministic switch from System 2 to System 1. A more defensible interpretation is that rapid cues and organizational hierarchy can increase reliance on heuristic judgment when time and working memory are constrained [13], [14]. At the same time, Truth-Default Theory predicts that the caller may remain provisionally trusted until a strong inconsistency or sensitive request triggers suspicion [4].

6.2.3 Authorization-boundary analysis

The relevant transition is from "this may be the executive" to "this executive request is authorized to alter the payment process." The synthetic voice contributes primarily to the first judgment. The threat actor then attempts to collapse the distinction by making procedural delay appear unreasonable or professionally costly. Security impact occurs

only if the DM allows perceived executive identity to substitute for the required approval path.

This mechanism can be described as pressure toward compliance. The caller’s apparent seniority changes the perceived consequences of hesitation: verifying the request may feel like challenging authority or obstructing an urgent business objective. Under time pressure, that framing can increase reliance on readily available cues [14], while a pre-existing truth-default can postpone active deception detection until a sufficiently diagnostic trigger appears [4]. The model nevertheless avoids assuming cognitive failure. A DM who treats policy adherence as part of professional authority can reinterpret the same pressure as a reason to insist on the formal approval path rather than bypass it.

The strongest failure condition is procedural confidence. If the DM understands that adherence to the approval rule is part of their authority and that executive urgency does not invalidate the requirement, the caller’s hierarchy loses leverage. A mandatory second approver, independently initiated callback, or written authorization can terminate the path without requiring the DM to determine whether the voice is synthetic.

6.3 Scenario 2: Retail-Customer Impersonation

6.3.1 Scenario setup

The second scenario places the Decision-Maker in the suspicious-transaction review function of Organization X Financial Services. The Retail Customer Identity is an account holder whose payment has been stopped as suspicious. The customer is also a peripheral social acquaintance of the DM: they occasionally encounter one another through a shared gym, hobby club, or mutual social circle, but they do not maintain a close friendship. Threat Actor A presents as the customer and requests reauthorization. The assumed reference material comes primarily from social media and is therefore more informal and potentially more emotionally varied than the executive’s public recordings, although recording quality may be less consistent.

The normal authentication path requires an approved channel such as in-app confirmation, formal account authentication, or in-person verification. The prior acquaintance does not alter these requirements, but it introduces an additional informal identity cue into the interaction. The scenario is deliberately independent of any specific STS implementation because the local RVC and Seed-VC validation did not establish operational suitability.

6.3.2 Influence structure and identity presentation

The trust relationship is procedural and interpersonal rather than hierarchical, but it is reinforced by limited prior familiarity. The caller has no authority over the DM. However, the claimed customer is not a complete stranger: the DM has encountered her occasionally in a shared social setting and therefore possesses a weak pre-existing identity representation. The relationship is sufficiently distant that the DM is unlikely to know her private circumstances or conversational behavior in detail.

This limited familiarity is analytically important. A recognizable name, broadly familiar voice, and plausible personal context can reinforce the initial identity claim, while the social distance leaves fewer opportunities for the DM to detect subtle inconsistencies in vocabulary, rhythm, or emotional behavior. Familiarity is therefore not treated as authentication evidence, but as an additional contextual factor affecting perceived legitimacy.

The presentation emphasizes vulnerability: the customer appears overwhelmed, desperate, or near tears, possibly while managing a crying child or another plausible source of distraction. These cues do not authenticate identity, but they can make the urgency and conversational imperfections coherent with the claimed circumstances.

The social mechanism is distress-oriented. Empathic concern can bias prosocial decision-making [35], and recent large-scale vishing research found the highest susceptibility in an emotionally personalized relative-in-distress condition [3]. The present scenario is not identical to that experiment and therefore does not inherit its compliance rate. The relevant inference is narrower: distress can change the recipient's decision frame by making procedural delay appear to worsen the situation of someone who seems legitimate and dependent on assistance.

6.3.3 Authorization-boundary analysis

The caller attempts to reposition the DM from security enforcer to crisis facilitator. Target-like speech, limited prior familiarity, account context, and emotional coherence can mutually reinforce the claimed identity and help maintain the recipient's truth-default, while the service relationship encourages assistance. Yet the protected decision remains formally unchanged: a stopped transaction should be reauthorized only after approved customer authentication. The acoustic identity cue and prior social familiarity therefore become dangerous only if they are permitted to substitute for the formal authentication step.

This mechanism draws the DM toward assistance rather than pushing through hierarchy. The analytical risk is a change in the perceived cost of enforcing procedure: continued verification can appear to prolong the distress of someone the DM already recognizes

socially, while rapid resolution can appear both compassionate and consistent with ordinary customer service. That pressure becomes security-relevant only when organizational discretion is allowed to substitute for authenticated evidence.

The strongest failure condition is authentication that remains independent of both emotional state and prior familiarity. Mandatory in-app confirmation, a verified callback, or account-specific verification prevents social recognition or distress from becoming authorization evidence. The interaction can also fail when the emotional performance appears exaggerated, contextual audio is inconsistent, or the caller’s vocabulary, behavior, or personal context conflicts with what the DM remembers from previous encounters.

6.4 Cross-Scenario Comparison

Tab. 4 Comparison of the two standardized influence structures.

Dimension	Executive-identity scenario	Retail-customer scenario
Primary trust basis	Hierarchy and executive authority	Customer legitimacy, limited prior familiarity, and service relationship
Influence direction	Pressure toward compliance	Draw toward assistance
Dominant affect	Impatience, forcefulness, urgency	Distress, vulnerability, overwhelm
Reference-audio profile	Social media, interviews, press/news, and public appearances	Primarily social-media recordings
Primary cognitive risk	Reduced deliberation under authority and time pressure	Prosocial bias and reluctance to prolong distress
Likely suspicion triggers	Unusual request, policy exception, or wording/style mismatch	Authentication avoidance, account inconsistency, or exaggerated distress
Protected process	Payment approval	Transaction reauthorization or account information
Strongest procedural barrier	Written or secondary executive authorization	Independent customer authentication

The comparison demonstrates that the same synthetic identity signal can participate in opposite interpersonal dynamics. In the executive scenario, pressure is directed at the DM through authority and urgency. In the customer scenario, pressure is generated by drawing the DM toward assistance. This distinction is analytically useful because it shows that voice cloning does not encode one universal manipulation strategy; it supplies an identity cue that can be embedded within different social relationships. Despite this difference, both scenarios converge on the same security boundary. The DM first evaluates

whether the caller may be the claimed person and then whether the request is permitted to alter a protected process. Voice cloning affects the first judgment more directly than the second. Organizational controls are robust when they preserve that separation. The contrast can therefore be formalized as two influence directions applied to the same authorization architecture. In Scenario 1, authority and urgency push the DM toward immediate compliance by increasing the perceived cost of resistance. In Scenario 2, distress and service expectations draw the DM toward assistance by increasing the perceived cost of delay. Both mechanisms attempt to transform an informal legitimacy judgment into authorization-relevant trust, but neither changes the formal evidence that should be required. This is the central comparative result: heterogeneous social mechanisms can converge on an identical technical control point, which is why authorization design is a more general defense than attempting to classify every persuasive cue.

7 Supplementary Validation of Technical Boundary Conditions

7.1 Evaluation Design

This chapter provides contextual validation for selected technical assumptions rather than a co-equal empirical contribution. All tests were local, German-language, and based on one consenting test speaker who was unrelated to the scenario identities. The workstation used an Intel Core i9-11950H, 64 GB RAM, and an NVIDIA RTX A5000 Laptop GPU with 16 GB VRAM. XTTS-v2 was the primary TTS system; RVC v2 and Seed-VC were supplementary VC checks. No listener study, English control, biometric evaluation, or real-time STS call was conducted.

7.2 XTTS-v2 Findings

From approximately 2 hours and 39 minutes of available consented German-Austrian speech, a 22.8-second reference was selected for the final XTTS speaker representation. Neutral German synthesis was intelligible and recognizably target-like, while a text-normalization stage improved numbers and abbreviations. The main limitation was

emotional and prosodic control: forceful, angry, desperate, or near-tearful generations were substantially less reliable and often sounded thinner or more synthetic.

Under stable conditions, complete-WAV generation took approximately 0.9-1.1 s for five words, 1.9-2.4 s for 13 words, and 4.1-4.8 s for 27 words (RTF approximately 0.40-0.47). Synthesis was therefore faster than playback, but these figures exclude comprehension, response formulation, text entry, and playback; occasional duration anomalies also required regeneration.

7.3 Speech-to-Speech Findings

RVC produced clear, stable, intelligible German speech, but perceptual target identity remained inadequate even after a larger 800-epoch trial despite high embedding-similarity scores. The mismatch is consistent with research on incomplete timbre/prosody disentanglement and the limited coverage of dynamic identity cues in common speaker embeddings [19], [20]. Seed-VC, a zero-shot diffusion-transformer approach [36], improved resemblance but introduced missing words, substitutions, malformed endings, and repetition loops; content instability therefore prevented conversational testing.

8 Discussion and Conclusion

8.1 Main Findings

The thesis finds that voice-cloning-based telephone impersonation is best modeled as a socio-technical authorization threat. Synthetic speech changes the plausibility of a caller identity, but the security impact of the call depends on a larger chain: a relevant recipient must be reached, context must align, the interaction must remain credible, suspicion must not trigger effective verification, and the DM must possess authority over a protected process. This result is consistent with recent evidence that persuasiveness and scenario context can explain susceptibility more strongly than voice human-likeness alone [3]. The threat model also clarifies why voice realism is neither necessary nor sufficient. Barrington et al. and Lavan et al. show that modern clones can be difficult to distinguish from authentic speech [1], [2], while familiarity and prosodic mismatch can improve a listener’s ability to reject a clone as the same person [17]. The local XTTS validation reaches the

same conceptual distinction from a different direction: neutral speaker resemblance was achievable, whereas strong emotional performance was less reliable. The supplementary VC observations add a second measurement lesson. Automatic speaker similarity and human identity perception are related but not interchangeable: embeddings can represent static timbral properties while omitting dynamic features that listeners use to recognize a person [19]. The RVC result, in which objective similarity increased while perceptual identity remained inadequate, should therefore be interpreted as a local example of a broader evaluation problem rather than as evidence that one metric or architecture is intrinsically invalid. The two scenarios demonstrate that the identity cue can operate through different influence structures. Hierarchical pressure attempts to make procedural resistance feel professionally costly; distress-oriented influence attempts to make procedural resistance feel unsympathetic or obstructive. In both cases, Truth-Default Theory provides a useful account of why the call may remain plausible until a request, inconsistency, or policy requirement activates suspicion [4].

8.2 Answers to the Research Questions

8.2.1 Main research question

Voice cloning alters perceived caller legitimacy by supplying a target-like paralinguistic identity signal that can reinforce caller ID, timing, contextual knowledge, and the expected social relationship. That signal becomes security-relevant only when the interaction remains coherent enough to survive suspicion and when the recipient permits the informal identity judgment to influence a protected authorization decision. Technical quality therefore interacts with contextual plausibility, trust structure, conversational timing, familiarity, and verification procedure rather than operating as an autonomous attack mechanism.

8.2.2 Sub-question 1

The principal assets are funds, confidential information, account information, transaction state, and the integrity of approval or authentication workflows. The principal adversary capability is not merely voice synthesis but the combination of target-speaker material, contextual knowledge, access to a relevant DM, and an interaction method capable of maintaining plausibility. The decisive trust boundary is the human authorization boundary.

8.2.3 Sub-question 2

The attack tree captures logical prerequisites, while the seven-stage lifecycle orders them over time. Identity success, interaction success, and security-impact success are deliberately separated. The lifecycle further integrates truth-default maintenance and suspicion triggers into the stage in which influence and verification occur.

8.2.4 Sub-question 3

Contextual alignment determines whether identity, timing, location, and request fit together. Authority and urgency can reduce deliberative opportunity; distress can increase prosocial pressure; familiarity can either support the initial identity prior or expose speaker-specific mismatch; and environmental audio can strengthen or weaken contextual coherence. None of these factors automatically causes compliance. Their security relevance depends on whether they delay or suppress independent verification.

8.2.5 Sub-question 4

The main technical constraints are conversational response preparation, emotional/prosodic fidelity, target-speaker resemblance, and content integrity. The main procedural constraints are substantially stronger: independent callbacks, written or in-app approval, dual control, step-up authentication, and employee authority to refuse irregular requests interrupt the attack path regardless of acoustic plausibility.

8.3 Contribution and Defensive Implications

The academic contribution is a bounded integration rather than a claim of theoretical firstness. Prior work already demonstrates multistage social engineering [5], synthetic voice risk [37], and decision-layer analysis of generative fraud [34]. This thesis contributes a call-specific cybersecurity model that links target-speaker audio, synthetic identity presentation, truth-default and suspicion, contrasting influence structures, independent verification, and protected organizational authorization. The attack tree explains logical dependence; the lifecycle explains temporal development; and the scenarios demonstrate how the same technical cue operates under different trust relationships.

For cybersecurity practice, the model argues against treating familiar voice or caller ID as sufficient evidence for sensitive actions. Deepfake detection remains valuable, but ASVspoofer 5 demonstrates that robustness changes under adversarial and channel conditions [31]. Higher-assurance processes should instead use authenticated channels and

phishing-resistant mechanisms that do not depend on correctly classifying the audio [6]. The design objective is not to make employees perfect deepfake detectors; it is to make a mistaken acoustic judgment insufficient to authorize material impact.

This distinction separates detection from authorization. A detector can provide useful risk evidence and can justify escalation, but its output remains probabilistic and exposed to distribution shift, codecs, and adversarial manipulation [31]. Conversely, NIST defines phishing resistance through cryptographic authentication that does not depend on the claimant’s vigilance and explicitly prohibits voice-based biometric comparison [6]. Applied to the modeled scenarios, the strongest control is therefore not an endpoint judgment that a voice is genuine or fake; it is an independently authenticated approval path whose security property remains valid even when the human legitimacy judgment is wrong.

The threat model consequently supports a layered control interpretation. Perceptual awareness and deepfake detection operate at the identity-presentation boundary and can raise suspicion; organizational policy operates at the human authorization boundary and determines when verification is mandatory; cryptographic or independently authenticated approval protects the process boundary itself. These layers are complementary rather than interchangeable. Detection can reduce exposure, but authorization controls determine whether a false acceptance becomes material impact. This allocation of responsibility is also more robust to technological change: improvements in voice realism, emotional expressiveness, or conversational latency may make the upstream deception harder to recognize without changing the downstream requirement for independent authorization.

8.4 Limitations

The scenarios are analytical and do not estimate prevalence or behavioral success rates. They simplify organizational structure to two protected processes and two influence mechanisms. The literature base is interdisciplinary and rapidly evolving; several 2026 sources used to position the research gap remain preprints and should not be given the evidential weight of peer-reviewed studies. The threat model also excludes autonomous agents, video, messaging, voice-biometric attacks, and direct system compromise.

The supplementary technical validation is limited to XTTS-v2, RVC, and Seed-VC on one workstation, one consenting target speaker, and German-language material. No English control, population listener study, biometric experiment, or real-time STS conversation test was performed. Existing bilingual research suggests that language alone is unlikely to explain the observed STS limitations, so the results cannot be attributed specifically to German [27]. The local findings therefore characterize implementation-specific boundaries rather than technology-wide performance.

8.5 Conclusion

Modern synthetic speech weakens the assumption that a familiar voice is reliable evidence of identity. Human listeners can accept high-quality clones as real or identity-consistent, and large-scale listening research shows that persuasive context can convert this capability into meaningful susceptibility [1], [3]. Yet the same evidence also shows that perfect acoustic fidelity is not the dominant variable in every interaction.

The threat becomes significant when synthetic identity, contextual knowledge, influence, timing, recipient authority, and weak verification align. The executive and retail-customer scenarios demonstrate that this alignment can be produced through different social mechanisms, but both converge on the same authorization boundary. Voice cloning should therefore be understood not as an autonomous cyberattack, but as a paralinguistic trust-establishing vector within an existing social-engineering framework.

This framing also makes the analysis durable beyond the specific models evaluated in Chapter 7. Future systems may reduce synthesis delay, reproduce emotional prosody more faithfully, or achieve more consistent zero-shot conversion. Such advances would increase the plausibility and flexibility of the identity-presentation stage, but they would not alter the structure of the protected decision. The central defensive requirement therefore survives technical progress: voice, caller ID, contextual knowledge, and emotional coherence may support a claim of identity, but none should constitute sufficient evidence to authorize a high-impact action.

The strongest defensive implication is structural. Sensitive processes should be engineered so that subjective identity judgments - however convincing the voice, caller ID, context, or emotion - cannot independently authorize a protected change. When independent, high-assurance authentication and approval remain mandatory, the synthetic voice may influence perception without determining security outcome.

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Overview of AI technologies used

Technology	Use
OpenAI GPT and Google Gemini	Refinement of academic writing style. Generation of LaTeX syntax and formatting. Proofreading for grammatical, spelling, and punctuation errors. Explanation of complex reference papers for personal comprehension. Assistance with Voice generation/conversion for the practical demonstrations.

List of Figures

1	System context and principal trust boundaries.	14
2	Attack tree for unauthorized protected action through voice-based impersonation.	17
3	Seven-stage lifecycle of voice-cloning-based telephone impersonation.	19

List of Tables

1	Core elements of the threat model.	13
2	STRIDE categories and their application in the threat model.	15
3	Representative failure conditions and defensive controls across the lifecycle.	19
4	Comparison of the two standardized influence structures.	25